

Cell Space

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ABSTRACT

“Cell Space” is a participatory mixed reality performance blending collocated mixed reality, neurofeedback, and contact improvisation. Each dancer inhabits a dynamic Voronoi-generated “cell” that displays real-time biofeedback signals—including heartbeat and pressure levels—as fluid boundaries. Through these interactions, “Cell Space” fosters deep intercorporeal connections, inviting participants and observers to explore the shifting boundaries between individual and collective intentionality while challenging conventional notions of embodied boundaries and dynamic spatial negotiation.

Index Terms: Mixed Reality, Neurofeedback, Contact Improvisation, Interactive Performance Art, Intercorporeality

1 PROJECT URL

<https://reality.design/project/cell-space>

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2 DESCRIPTION OF ARTWORK

2.1 Thematic Statement

In the post-COVID-19 era, we’ve become more attuned to social distancing and personal boundaries. The skin may no longer represent the true physical limit of the human body. During the pandemic, with viruses suspended in the air between people, we must question: What are the actual boundaries of the human body? Would visualizing these boundaries enhance our awareness of the physical and biological space between ourselves and others?

“Cell Space” acts as a reflective mirror for potential machine-mediated futures in an increasingly mixed-reality society. It heightens participants’ awareness of their physical boundaries, illuminating the divisions between body, mind, and environment. This contributes to a world where technology and humanity converge in solidarity.

2.2 Artwork

Cell Space is a participatory mixed reality art performance that merges collocated mixed reality, neurofeedback, and contact improvisation, redefining the interaction between spectators and participants by intertwining the physical with the digital, and the organic with the technological. Integrating augmented reality, neurofeedback, and contact improvisation, Cell Space fosters deep intercorporeal connections, blurring the lines between individual and collective intentionality. In contemporary media art, the boundaries between audience and performer, the physical and virtual, and the organic and technological are becoming increasingly blurred. Reflecting this evolving landscape, “Cell Space” invites participants to

transcend traditional roles and boundaries, reimagining themselves as dynamic cells within a shared bio-digital ecosystem. Inspired by McLuhan's idea of media as an extension of the human body, Jess Rowland's emphasis on internal awareness and the vision presented in *The AR Art Manifesto* of "we becoming the Media", this artwork creates a living, evolving entity through contact improvisation. This dynamic ecosystem, resonating with Haraway's concept of the fluid cyborg, enables a new form of intercorporeality, combining physical touch and bio-data within a digitally augmented reality space. It advances the discourse in immersive and interactive arts, presenting a bodily interface that promotes intercorporeal interactions and responds to both individual and collective intentions of its participants.

At the core of *Cell Space* lies a deep exploration of contact improvisation, a dance form that moves through intangible elements such as imagination, emotions, and connections with others and the environment. These elements, while essential, can be elusive, particularly for participants seeking a more profound connection with themselves and others. Through *Cell Space*, neurofeedback technology captures bio-data and visualizes these intangibles, while augmented reality enhances the intercorporeal experience, facilitating a seamless "perception-action loop between self and other". MR and neurofeedback serve as perceptual amplifiers, balancing immersion and distraction to allow participants to fully engage with their movements and emotions. The three phases—Opening (Division), Developing (Differentiation), and Closing (Death)—mirror the life cycle of cells and the fluidity of digital entities. In the Opening phase, participants explore contact through simple duet exercises, symbolizing the start of a bio-digital life cycle.

2.3 Implementation

Developed as an iPhone app using Unity3D, and designed for the HoloKit X MR headset and EEG headbands, "*Cell Space*" creates a co-located MR space where participants, categorized as Immersants, Spectators, and Bystanders, experience a blend of the physical and digital. This minimalist yet immersive environment, encourages users to explore the nuanced interplay between their digital and physical identities, promoting a reflective look at our digitally mediated existence. Through this, *Cell Space* explores the evolution of human-machine-space symbiosis and poses critical reflections on societal norms and the complexities of our interconnected lives.

2.4 Participant Roles

Cell Space accommodates different levels of participation, defining three primary roles: Immersants, Spectators, and Bystanders. Immersants are dancers wearing HoloKit MR headsets, which connect them to the performance space and other dancers with minimal latency, while displaying augmented cell structures that reflect their proximity and mental states. They also wear Flowtime biosensors to translate their heart rate and EEG signals into visual elements within the space. Spectators use AR headsets or handheld devices to observe the performance in real-time, seeing the augmented cell structures and potentially transitioning from passive viewers to active participants by entering the performing field. Bystanders, without devices, engage through a monitor screen, gaining a third-person view of the immersive experience, allowing them to appreciate the performance's dynamic interactions and movements.

2.5 Relevance to the Theme

In alignment with the IEEE VR theme of questioning the nature of reality, *Cell Space* explores the potential for profound unity through shared bio-digital experiences. By co-creating an interconnected and dynamic ecosystem, the artwork embodies the collective beauty and creation central to the theme. It transcends human-machine interaction, evolving into a human-machine-space symbiosis. Participants

contribute to a shared tapestry where individual experiences resonate within the collective, embodying the future of interconnection: a world where boundaries dissolve and unity emerges from the interplay of physical and digital realms.

Ultimately, *Cell Space* serves as a mirror to the future of our interconnected existence. It encourages participants to explore a transformative landscape that reflects the dynamic relationships between their physical and digital identities and their surroundings, fostering collaboration and symbiosis in a world increasingly shaped by technological and human connections.

3 ARTIST INFORMATION

Rem RunGu Lin is a digital artist, researcher, and technologist with a master's degree from MIT, currently pursuing a PhD in Computational Media and Art at HKUST (GZ). His work explores the intersections of embodied interaction, brainmedia, and media theory, blending research with creative practice. His current research focuses on three key areas: Media as Art, exploring how media theory can be applied to art and design to augment creative practices; Human as Media, investigating the entanglement of body and mind and how this relationship can be expressed through digital art; and Space as Media, extending the concept of human-machine-space symbiosis to create immersive, interactive bio-digital spatial entities. His research has been published in SIGGRAPH, LEONARDO, SIGGRAPH Asia, VCIBA, IEEE VIS AP, CHI PLAY, ISEA, and VINCI. His artworks have been exhibited at ISEA Creative Work, SIGGRAPH ASIA XR, Ars Electronica, IEEE VIS AP Art Gallery, DIGITAL ART CHINA, etc..

Botao 'Amber' Hu is a social computing researcher and experiential futures designer. He directs Reality Design Lab (<https://reality.design>), an independent interdisciplinary research and design lab exploring the intersection of soma design (bodily play), speculative design (artificial life), spatial computing (mixed reality), and social computing (protocol science, public-interest cryptography, decentralized AI governance). His primary focus is designing socially experiential futures using mixed reality as the main medium. He also serves as a visiting lecturer at the China Academy of Art. His works have been featured at top conferences such as SIGGRAPH, CSCW, CHI, UbiComp, WWW, TEI, ISEA, IEEEVR, IEEEVIS, ISMAR, Ars Electronica, SXSW, and TEDx and have received accolades including the SIGGRAPH Best in Show, CHI Best Interactivity, Webby, Red Dot, iF Design, Good Design, A' Design, Core77 Design Award, and grants from the Ethereum Foundation. He holds a bachelor's degree in computer science from Tsinghua University and a master's degree in computer science with AI concentration from Stanford University.

Koo Yongen Ke is a new media artist and co-founder of Funtheory/Befun Digital Lab. She holds a Bachelor of Music from Jinan University, with a major in composition and electronic music production and is accepted to PhD in Computational Media and Art at HKUST (GZ). As the director of the Guangzhou Modern Music Association, she is renowned for her work in immersive interactive video creation. Koo has collaborated extensively across disciplines including dance, drama, and music, notably with Tan Dun on projects like the "Buddha Passion" symphony at MGM Cotai and the "Tan Dun WE-Festival" at the Hong Kong Cultural Centre.